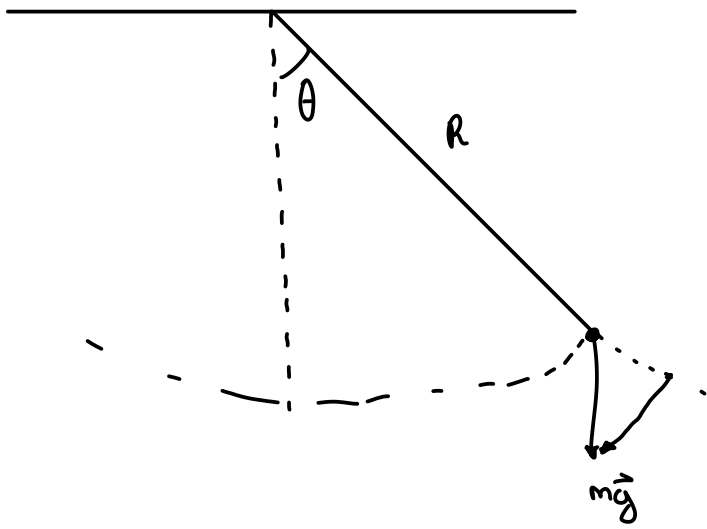
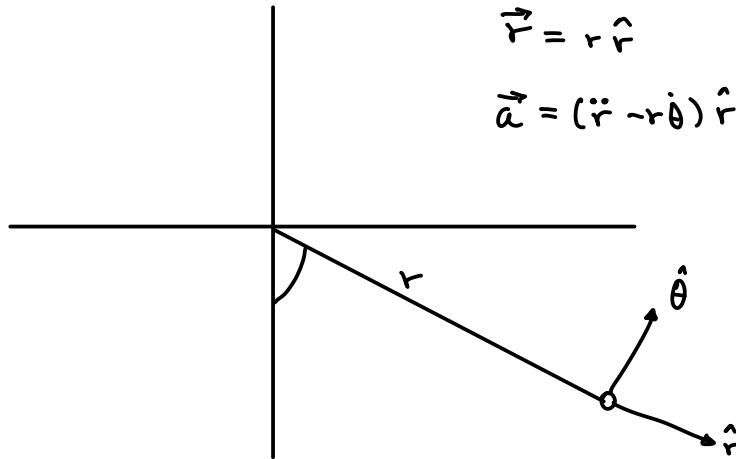




Tenemos...



$$\vec{r} = r \hat{r}$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2) \hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta}) \hat{\theta}$$

$$r = R$$

$$\dot{r} = 0$$

$$\ddot{r} = 0$$

$$\ddot{\theta} = -\frac{g}{R} \sin \theta$$

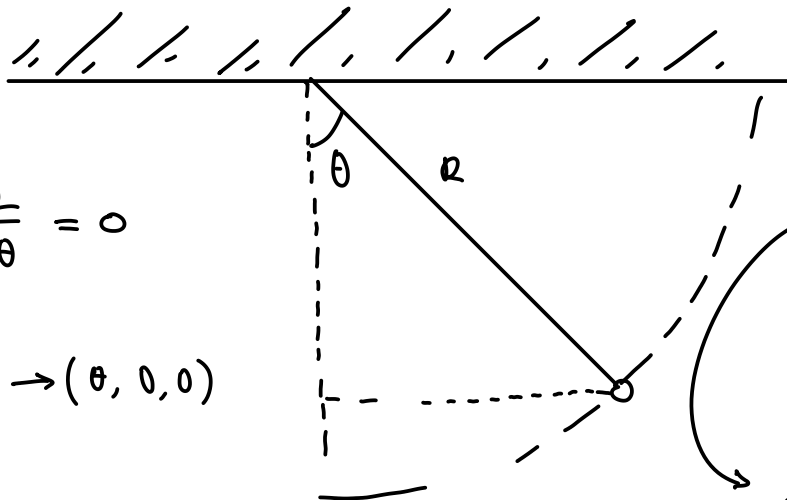
$$\vec{a} = (\ddot{r} - r\dot{\theta}^2) \hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta}) \hat{\theta}$$

Tenemos

Lagrangiano = $T - U$; coordenada generalizada

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0$$

$$(\theta, \dot{\theta}, R) \rightarrow (\theta, \dot{\theta}, 0)$$



• masa obligada a moverse sobre un círculo

→ construido a moverse en un círculo

Punto por:

$$x^2 + y^2 = R^2$$

$$x^2 + y^2 + z^2 = R^2 ; z = 0$$

$$x = R \sin \theta$$

$$y = -R \cos \theta$$

$$f(x, y) = x^2 + y^2 - R^2 = 0$$

Energía cinética

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

$$\left. \begin{aligned} \dot{x} &= R \dot{\theta} \cos \theta \\ \dot{y} &= R \dot{\theta} \sin \theta \end{aligned} \right\} T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) = \frac{1}{2} m (R^2 \dot{\theta}^2 \cos^2 \theta + R^2 \dot{\theta}^2 \sin^2 \theta)$$

$$\vdots$$

$$= \frac{1}{2} m R^2 \dot{\theta}^2$$

• El péndulo tiene un grado de libertad

$$q = \theta$$

Potencial

$$U = mgy \Rightarrow U = mg(-R \cos \theta) = U = -mgR \cos \theta$$

Tenemos...

$$L(R, \theta, \dot{\theta}, t) = \frac{1}{2} m R^2 \dot{\theta}^2 - [-mgR \cos \theta] = \frac{1}{2} m R^2 \dot{\theta}^2 + mgR \cos \theta$$

E-L.

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0 \Rightarrow \frac{d}{dt} (m R^2 \dot{\theta}) + mgR \sin \theta \overset{\frac{d\theta}{d\theta} = 1}{\dot{\theta}} = 0$$

$$\Rightarrow m R^2 \ddot{\theta} + mgR \sin \theta \dot{\theta} = 0$$

$$\Rightarrow R^2 \ddot{\theta} = -gR \sin \theta \dot{\theta}$$

$$\Rightarrow \ddot{\theta} = -\frac{g}{R} \sin \theta \dot{\theta}$$

Hamiltoniano... (no depende de las velocidades)

Tenemos

$$H(R, \theta, p_\theta, t) = \sum_{i=1}^n \frac{\partial L}{\partial \dot{\theta}} \dot{\theta} - L(R, \theta, \dot{\theta}, t)$$

$$\vdots$$

$$= m R^2 \dot{\theta}^2 - \left[\frac{1}{2} m R^2 \dot{\theta}^2 + mgR \cos \theta \right]$$

$$\frac{\partial L}{\partial \dot{\theta}} = p_\theta = m R^2 \dot{\theta}$$

$$\dot{\theta} = \frac{p_\theta}{m R^2}$$

$$= mR^2 \dot{\theta}^2 - \frac{1}{2} mR^2 \dot{\theta}^2 - mgR \cos \theta$$

$$= \frac{1}{2} mR^2 \dot{\theta}^2 - mgR \cos \theta$$

$$= \frac{1}{2} mR^2 \left[\frac{p_\theta^2}{(mR^2)^2} \right] - mgR \cos \theta$$

$$H(R, \theta, p_\theta) = \frac{1}{2} \frac{p_\theta^2}{mR^2} - mgR \cos \theta$$

Ecuaciones canónicas de Hamilton...

$$\left. \begin{aligned} \dot{p}_i &= - \frac{\partial H}{\partial q_i} \\ \dot{q}_i &= \frac{\partial H}{\partial p_i} \end{aligned} \right\} \begin{array}{l} 2n \text{ ecuaciones diferenciales de} \\ \text{primer orden.} \end{array}$$

$$\dot{p}_\theta = - \left[\frac{\partial H}{\partial \theta} \right] = - \left[mgR \sin \theta \right] \Rightarrow \dot{p}_\theta = - mgR \sin \theta$$

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{mR}$$

Tenemos

$$\left. \begin{aligned} \dot{p}_\theta &= -mgR \sin \theta \\ \dot{\theta} &= \frac{p_\theta}{mR} \end{aligned} \right\} \begin{aligned} \ddot{\theta} &= \frac{\dot{p}_\theta}{mR} = - \frac{mgR \sin \theta}{mR} \\ \ddot{\theta} &= - \frac{g}{R} \sin \theta \end{aligned}$$